

Money Laundering Detection with Multi-Aggregation Custom Edge GIN - Supplementary Material

The data

This study builds upon the datasets and methodological framework established by (Silva, Correia, and Maziero 2023). Their research utilized data generated by the financial transaction simulation tool “AMLSim”, developed by IBM (Weber et al. 2018). AMLSIm extends the capabilities of the PaySim architecture and was designed to simulate transactional patterns characteristic of money laundering processes. This simulator stands out for its ability to fine-tune a wide range of parameters, such as the duration of the simulation in time steps (days), transaction value ranges, and class imbalance ratio, among other statistical variables.

In their study, Silva, Correia, and Maziero (2023) generated a dataset representing daily transaction activities for the year 2020, encompassing a total of 365 time steps. This dataset includes account identifiers, predetermined suspicion indicators, details on the sending and receiving accounts, and transaction types. The main elements for their analysis included the transaction value (*“base_amt”*), transaction timestamp (*“tran_timestamp”*), and transaction type (*“tx_type”*), along with flags indicating potential illicit activities (*“is_sar”*), and data on the source or destination accounts (*“orig_acct”*, *“bene_acct”*). These elements were used to construct the transaction graphs, supplemented by initial deposit information (*“initial_deposit”*) and transaction behaviour categorizations (*“tx_behavior_id”*) for each account. That dataset was divided into several portions with different class imbalance rates.

Table 1 summarizes the key elements of the datasets used in this study - number of nodes and edges in train/validation/test splits. Table 2 presents the class proportions for the different datasets.

Table 1: Dataset statistics

Dataset	# train nodes	# val nodes	# test nodes	# train edges	# val edges	# test edges
AMLSim 1/3	3889	1186	1256	1977	599	637
AMLSim 1/5	6338	2036	2133	3226	1039	1090
AMLSim 1/10	12330	4239	4182	6366	2184	2160
AMLSim 1/20	24283	7945	8036	12931	4222	4267

Table 2: Class proportions per dataset

Dataset	# legal	% legal	# illicit	% illicit
AMLSim 1/3	1317	0.666	660	0.334
AMLSim 1/5	2566	0.795	660	0.205
AMLSim 1/10	5706	0.896	660	0.104
AMLSim 1/20	12271	0.949	660	0.051

Hyperparameter optimization

While a traditional ablation study was not conducted, the contribution of each hyperparameter to model performance was systematically evaluated using the PED-ANOVA Importance. This method serves as a possible alternative to ablation studies, particularly in cases where the number of tuned parameters is large and their interactions form subspaces (e.g., the interplay between the number of layers, layer size, and aggregation methods per layer). By leveraging ANOVA-based calculations, PED-ANOVA quantifies how strongly each hyperparameter contributes to achieving performance above a baseline, which is determined by fitting Parzen estimators to the results of completed trials (Watanabe, Bansal, and Hutter 2023).

The feature importance analysis revealed that no single hyperparameter dominated performance across all datasets. Instead, the relative importance varied depending on dataset properties, particularly the class imbalance ratio and dataset size. For less-imbalanced cases (AMLSim 1/3 and 1/5), the number and size of convolutional layers, as well as aggregation functions, were relatively more influential, though their importance scores were comparable. In contrast, in highly imbalanced scenarios (AMLSim 1/10 and 1/20), embedding reduction modes became more critical, alongside aggregation functions.

This variability highlights the importance of adaptive hyperparameter tuning tailored to dataset-specific properties. The flexibility of the MAGIC architecture further supports effective performance under a wide range of configurations, including the use of multiple, learnable aggregations.

Sections below present details on subsequent aspects of the hyperparameter optimization results - jointly and individually for each dataset.

Hyperparameter importances

Figure 1 illustrates the relative importance of each hyperparameter across the different datasets. The importance scores were calculated using the PED-ANOVA method, which quantifies the contribution of each hyperparameter to the model’s performance above a baseline. The baseline was determined by fitting Parzen estimators to the results of completed trials.

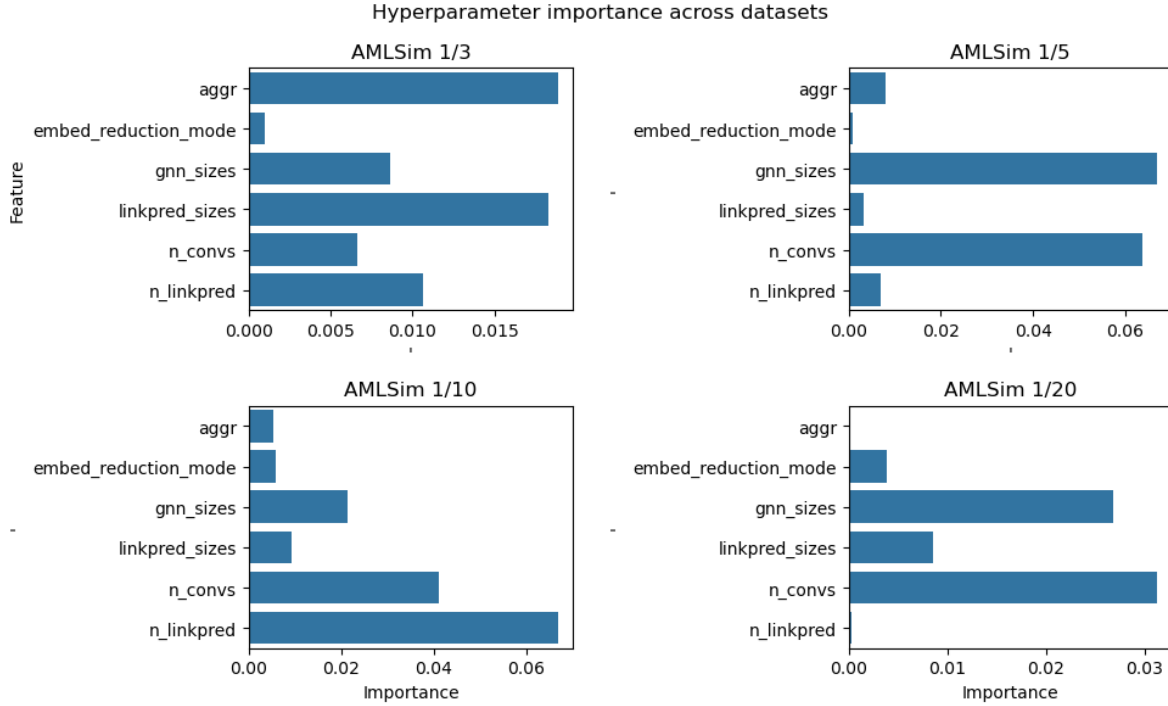


Figure 1: PED-ANOVA hyperparameter importances across datasets.

Subsequent tables Table 3, Table 4, Table 5, and Table 6 present the detailed results of the PED-ANOVA analysis for each dataset. The tables include the hyperparameter name, the importance F-score, and the corresponding p-value.

Table 3: Anova results for AMLSim 1/3.

Source	SS	DF	MS	F	p-unc	np2	Significance
Conv. Dim.	0	3	0	0	1	0	No
No. Conv. Layers	-0	3	-0	-0	1	-0	No
Aggr.	-0	2	-0	-0	1	-0	No
Embed Reduction	-0	1	-0	-0	1	-0	No
Conv. Dim. * No. Conv. Layers	0	9	0	0	1	0	No
Conv. Dim. * Aggr.	-0	6	-0	-0	1	-0	No
No. Conv. Layers * Aggr.	0	6	0	0	1	0	No
Conv. Dim. * Embed Reduction	-0	3	-0	-0	1	-0	No
No. Conv. Layers * Embed Reduction	-0	3	-0	-0	1	-0	No
Aggr. * Embed Reduction	0	2	0	0	1	0	No
Conv. Dim. * No. Conv. Layers * Aggr.	3.1215	18	0.1734	2.7434	0.03	0.7329	Yes
Conv. Dim. * No. Conv. Layers * Embed Reduction	1.5091	9	0.1677	2.6526	0.067	0.5701	No
Conv. Dim. * Aggr. * Embed Reduction	0.7294	6	0.1216	1.9232	0.1402	0.3906	No
No. Conv. Layers * Aggr. * Embed Reduction	2.2311	6	0.3719	5.8826	0.0022	0.6623	Yes
Conv. Dim. * No. Conv. Layers * Aggr. * Embed Reduction	2.2194	18	0.1233	1.9506	0.083	0.6611	No
Residual	1.1378	18	0.0632	nan	nan	nan	No

Table 4: Anova results for AMLSim 1/5.

Source	SS	DF	MS	F	p-unc	np2	Significance
Conv. Dim.	0	3	0	0	1	0	No
No. Conv. Layers	0	3	0	0	1	0	No
Aggr.	0	2	0	0	1	0	No
Embed Reduction	-0	1	-0	-0	1	-0	No
Conv. Dim. * No. Conv. Layers	-0	9	-0	-0	1	-0	No
Conv. Dim. * Aggr.	0	6	0	0	1	0	No
No. Conv. Layers * Aggr.	0	6	0	0	1	0	No
Conv. Dim. * Embed Reduction	-0	3	-0	-0	1	-0	No
No. Conv. Layers * Embed Reduction	-0	3	-0	-0	1	-0	No
Aggr. * Embed Reduction	-0	2	-0	-0	1	-0	No

Table 4: Anova results for AMLSim 1/5.

Source	SS	DF	MS	F	p-unc	np2	Significance
Conv. Dim. * No. Conv. Layers * Aggr.	15.0757	18	0.8375	54.3891	0	0.977	Yes
Conv. Dim. * No. Conv. Layers * Embed Reduction	0.7099	9	0.0789	5.1219	0.0074	0.6671	Yes
Conv. Dim. * Aggr. * Embed Reduction	0.8282	6	0.138	8.9633	0.0004	0.7004	Yes
No. Conv. Layers * Aggr. * Embed Reduction	0.3412	6	0.0569	3.6932	0.0264	0.4907	Yes
Conv. Dim. * No. Conv. Layers * Aggr. * Embed Reduction	2.7922	18	0.1551	10.0735	0	0.8874	Yes
Residual	0.3542	23	0.0154	nan	nan	nan	No

Table 5: Anova results for AMLSim 1/10.

Source	SS	DF	MS	F	p-unc	np2	Significance
Conv. Dim.	-0	3	-0	-0	1	-0	No
No. Conv. Layers	-0	3	-0	-0	1	-0	No
Aggr.	-0	2	-0	-0	1	-0	No
Embed Reduction	-0	1	-0	-0	1	-0	No
Conv. Dim. * No. Conv. Layers	0	9	0	0	1	0	No
Conv. Dim. * Aggr.	-0	6	-0	-0	1	-0	No
No. Conv. Layers * Aggr.	-0	6	-0	-0	1	-0	No
Conv. Dim. * Embed Reduction	0	3	0	0	1	0	No
No. Conv. Layers * Embed Reduction	-0	3	-0	-0	1	-0	No
Aggr. * Embed Reduction	0	2	0	0	1	0	No
Conv. Dim. * No. Conv. Layers * Aggr.	4.7261	18	0.2626	12.5896	0	0.9079	Yes
Conv. Dim. * No. Conv. Layers * Embed Reduction	2.9072	9	0.323	15.4888	0	0.8584	Yes
Conv. Dim. * Aggr. * Embed Reduction	1.2712	6	0.2119	10.1588	0.0002	0.726	Yes
No. Conv. Layers * Aggr. * Embed Reduction	1.8932	6	0.3155	15.1295	0	0.7979	Yes
Conv. Dim. * No. Conv. Layers * Aggr. * Embed Reduction	0.5755	18	0.032	1.533	0.1657	0.5454	No
Residual	0.4797	23	0.0209	nan	nan	nan	No

Table 6: Anova results for AMLSim 1/20.

Source	SS	DF	MS	F	p-unc	np2	Significance
Conv. Dim.	-0	3	-0	-0	1	-0	No
No. Conv. Layers	-0	3	-0	-0	1	-0	No
Aggr.	0	2	0	0	1	0	No
Embed Reduction	0	1	0	0	1	0	No
Conv. Dim. * No. Conv. Layers	0	9	0	0	1	0	No
Conv. Dim. * Aggr.	0	6	0	0	1	0	No
No. Conv. Layers * Aggr.	0	6	0	0	1	0	No
Conv. Dim. * Embed Reduction	-0	3	-0	-0	1	-0	No
No. Conv. Layers * Embed Reduction	-0	3	-0	-0	1	-0	No
Aggr. * Embed Reduction	-0	2	-0	-0	1	-0	No
Conv. Dim. * No. Conv. Layers * Aggr.	13.0613	18	0.7256	73.1902	0	0.9821	Yes
Conv. Dim. * No. Conv. Layers * Embed Reduction	0.9343	9	0.1038	10.4714	0	0.797	Yes
Conv. Dim. * Aggr. * Embed Reduction	0.0842	6	0.014	1.4153	0.2627	0.2614	No
No. Conv. Layers * Aggr. * Embed Reduction	0	6	0	0	1	0	No
Conv. Dim. * No. Conv. Layers * Aggr. * Embed Reduction	0.7184	18	0.0399	4.0256	0.0009	0.7512	Yes
Residual	0.2379	24	0.0099	nan	nan	nan	No

Complexity and quality

The tuning process revealed a consistent trade-off between model complexity and illicit F1 scores. Simpler models, characterized by fewer layers and smaller convolutional dimensions, matched or outperformed more complex variants every time. Figure 2 illustrates the relationship between model complexity and illicit F1 scores across all datasets. As a result, simpler architectures were selected whenever performance differences were negligible.

Figure 3 presents only the interaction between complexity (number of layers and their size) and illicit F1 scores. For every dataset simpler models exhibit similar or better performance than more complex ones.

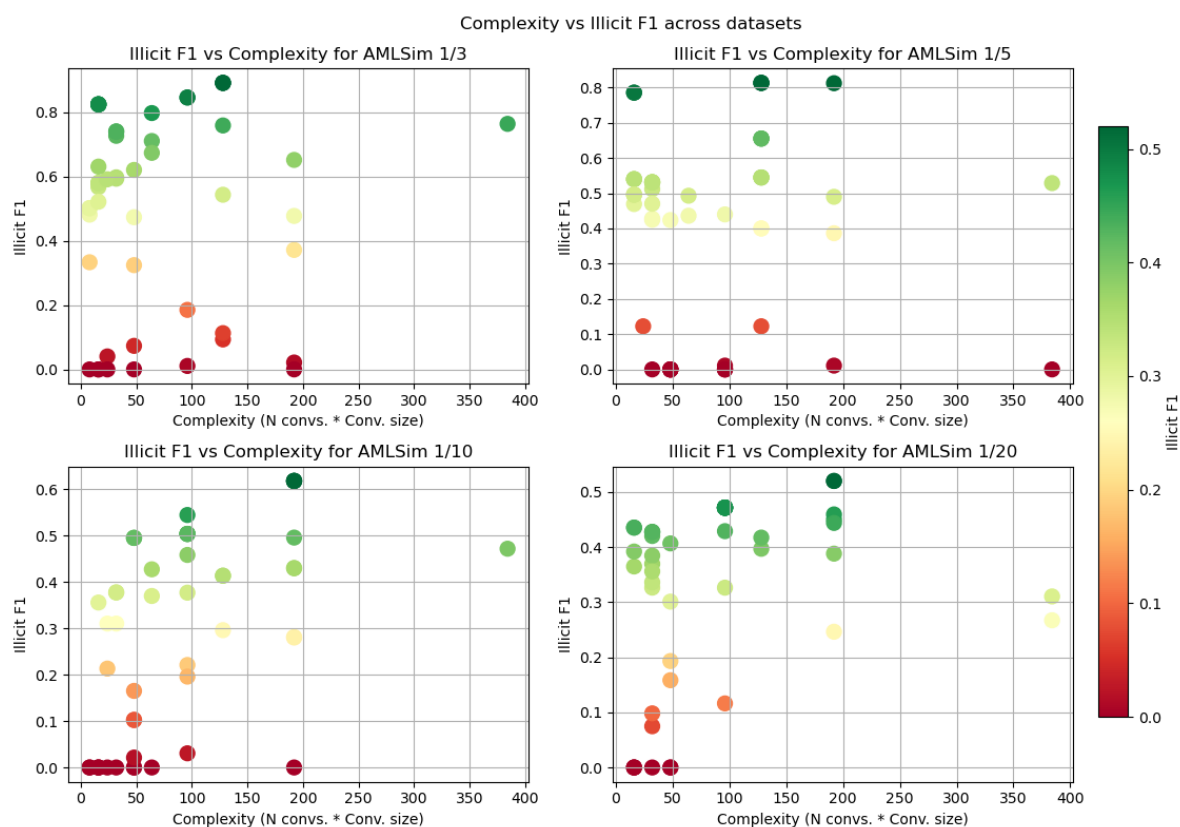


Figure 2: F1 score vs complexity across datasets.

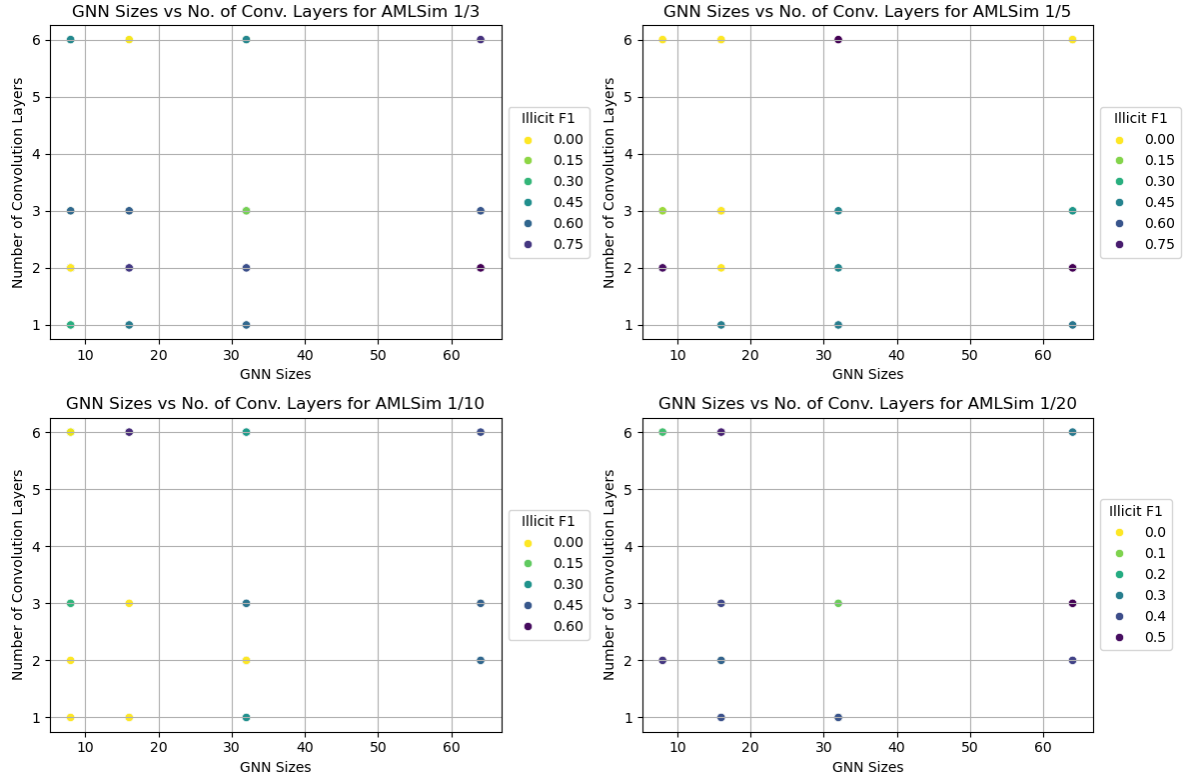


Figure 3: GNN sizes vs number of convolutions across datasets.

Model structure

This section presents the details of the final convolution layers selected for each dataset. Tables Table 7, Table 8, Table 9, and Table 10 summarize the final model configurations for the AMLSim 1/3, 1/5, 1/10, and 1/20 datasets, respectively.

Table 7: Model summary for AMLSim 1/3.

Layer	Input Shape	Output Shape	#Param
MAGICModel	[3889, 6], [2, 1977], [1977, 8]	[3889, 64]	27,000
(convs)ModuleList	–	–	27,000
(0)MAGICConv	[3889, 6], [2, 1977], [1977, 8]	[3889, 64]	5,815
(1)MAGICConv	[3889, 64], [2, 1977], [1977, 8]	[3889, 64]	21,185

Table 8: Model summary for AMLSim 1/5.

Layer	Input Shape	Output Shape	#Param
MAGICModel	[6338, 6], [2, 3226], [3226, 8]	[6338, 64]	18,040
(convs)ModuleList	–	–	18,040
(0)MAGICConv	[6338, 6], [2, 3226], [3226, 8]	[6338, 64]	5,047
(1)MAGICConv	[6338, 64], [2, 3226], [3226, 8]	[6338, 64]	12,993

Table 9: Model summary for AMLSim 1/10.

Layer	Input Shape	Output Shape	#Param
MAGICModel	[12330, 6], [2, 6366], [6366, 8]	[12330, 32]	5,272
(convs)ModuleList	–	–	5,272
(0)MAGICConv	[12330, 6], [2, 6366], [6366, 8]	[12330, 32]	855
(1)MAGICConv	[12330, 32], [2, 6366], [6366, 8]	[12330, 32]	4,417

Table 10: Model summary for AMLSim 1/20.

Layer	Input Shape	Output Shape	#Param
MAGICModel	[24283, 6], [2, 12931], [12931, 8]	[24283, 32]	4,056
(convs)ModuleList	–	–	4,056
(0)MAGICConv	[24283, 6], [2, 12931], [12931, 8]	[24283, 32]	663
(1)MAGICConv	[24283, 32], [2, 12931], [12931, 8]	[24283, 32]	3,393

References

- Silva, Ítalo Della Garza, Luiz Henrique Andrade Correia, and Erick Galani Maziero. 2023. “Graph Neural Networks Applied to Money Laundering Detection in Intelligent Information Systems.” *Proceedings of the XIX Brazilian Symposium on Information Systems*, May, 252–59. <https://doi.org/10.1145/3592813.3592912>.
- Watanabe, Shuhei, Archit Bansal, and Frank Hutter. 2023. “PED-ANOVA: Efficiently Quantifying Hyperparameter Importance in Arbitrary Subspaces.” *arXiv Preprint arXiv:2304.10255*.
- Weber, Mark, Jie Chen, T. Suzumura, A. Pareja, Tengfei Ma, H. Kanezashi, Tim Kaler, C. Leiserson, and T. Schardl. 2018. “Scalable Graph Learning for Anti-Money Laundering: A First Look.” *ArXiv: 1812.00076*.